WTW 158 2018 Worksheet 4 on Inverse Trig functions and Limits.

A. TEXTBOOK PROBLEMS

p66 no 63, 64, 68, 70, 71, 72, 75 p 82 no 1(a), 6 (a) - (i)
p 92 no 7, 9, 11, 17, 31, 34, 38

B. MORE PROBLEMS

1.1 Complete the definition:

$y = \cos x \Leftrightarrow$................ where ....

1.2 Sketch $y = \cos x, x \in [0,\pi]$ and $y = \arccos x$ on the same set of axes

2. Find the domain of the function $f(x) = \frac{1}{\arcsin(x)}$

3. Find the average rate of change of the function $f(x) = \ln x$ over the interval $[1,3]$

4. Does the function $f(x) = \frac{x^2+2x-8}{x^2-5x+6}$ have a vertical asymptote(s)?

5. Determine $\lim\limits_{x\to 0} (\frac{1}{x} - \ln x)$

6. (i) $\lim\limits_{x\to 2^+} \frac{x}{2-x}$ (ii) $\lim\limits_{x\to 2^-} \frac{x+1}{x-2}$

C. TEST QUESTIONS

Question 1

For which value(s) of x is $\tan(\tan^{-1}x) = x$?

(Notation $\tan^{-1} = \arctan = bg\tan$)

(a)	$0 \leq x \leq \pi$	(b)	$-1 \leq x \leq 1$	(c)	$-\pi \leq x \leq \pi$
(d)	$-\frac{\pi}{2} < x < \frac{\pi}{2}$	(e)	$x \in \mathbb{R}$	(f)	None of these

Question 2

$$\lim_{x\to a} \frac{x^2 - 2ax + a^2}{x^2 - a^2} =$$

(a)	$\frac{1}{2a}$	(b)	$-\frac{1}{2a}$	(c)	$-\frac{1}{2a^2}$
(d)	$\frac{1}{2a^2}$	(e)	0	(f)	None of these

Question 3

Does the function $f(x) = \frac{|x+5|}{x+5}$ have a vertical asymptote at $x = -5$? Explain.

D YOUTUBE

Kahn Academy - Introduction to limits

SELECTED ANSWERS

A. p66 63. π 64. $\frac{\pi}{3}$ 68. $-\frac{\pi}{4}$ 70. $\frac{x}{\sqrt{1-x^2}}$ 71. $\frac{x}{\sqrt{1+x^2}}$ 72. $2x\sqrt{1-x^2}$ 75. Domain: $[-\frac{2}{3}, 0]$
Range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$ p82 6(a) (i) 4.42 m/s p92 7(a) -1 (b) -2 (c) DNE (d) 2 (e) 0 (f) DNE (g) 1 (h) 3 9. (a) $-\infty$ (b) ∞ (c) ∞ (d) $-\infty$ (e) ∞ 11. $\lim\limits_{x\to a} f(x)$ exists for $a \in \mathbb{R}\backslash\{-1\}$ 31. ∞
34. $-\infty$ 38. $-\infty$

B. 1.1 See textbook 2. $x \in [-1,1]\backslash\{0\}$ 3. $\frac{\ln 3}{2}$ 4. VA $x = 3$ 5. ∞ 6.(i) $-\infty$ (ii) $-\infty$

C. 1 (e) 2(e) 3. No, $\lim\limits_{x\to 5^+} f(x) = 1$, $\lim\limits_{x\to 5^-} f(x) = -1$

MTW 158 Worksheet 6

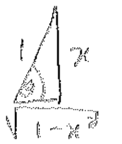
A. TEXTBOOK PROBLEMS p 66

63. $\cos^{-1}(-1) = \pi$

64. $\tan^{-1}\sqrt{3} = \dfrac{\pi}{3}$

68. $\arcsin\left(\sin \dfrac{5\pi}{4}\right) = -\dfrac{\pi}{4}$

70. $\tan(\underbrace{\sin^{-1}x}_{\theta}) = \dfrac{x}{\sqrt{1-x^2}}$



71. $\sin(\tan^{-1}x)$

$= \dfrac{x}{\sqrt{1+x^2}}$

72. $\sin(2\underbrace{\arccos x}_{\theta})$

$= 2\sin(\arccos x)\cdot\cos(\arccos x)$

$= 2\cdot\dfrac{\sqrt{1-x^2}}{1}\cdot x = 2x\sqrt{1-x^2}$

75. $y = \sin^{-1}(3x+1)$

Domain: $-1 \le 3x+1 \le 1$

$\Rightarrow -2 \le 3x \le 0$

$\Rightarrow -\dfrac{2}{3} \le x \le 0$

$\Rightarrow x \in [-2/3, 0]$

Range: $[-\pi/2, \pi/2]$

p 82

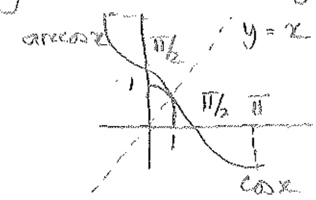
6(a)(i) Ave velocity on $[1, 2]$:

$\dfrac{10(2) - 1.86(2)^2 - \left(10(1) - 1.86(1)^2\right)}{2-1} = 4.42\,m/s$

B. MORE PROBLEMS

1.1 $y = \cos x \iff \cos y = x$ where $x \in [0, \pi]$

1.2



2. $f(x) = \dfrac{1}{\arcsin x}$

$x \in [-1, 1]$ but $\arcsin x \ne 0$

$\Rightarrow x \ne 0$

$\Rightarrow x \in [-1, 1] \setminus \{0\}$

3. Ave rate of change is

$\dfrac{\ln 3 - \ln 1}{3-1} = \dfrac{\ln 3}{2}$

4. $f(x) = \dfrac{(x+4)\cancel{(x-2)}}{\cancel{(x-2)}(x-3)}, \quad x \ne 2$

$= \dfrac{x+4}{x-3}$

$\lim_{x\to 3^+} \dfrac{x+4}{x-3} = \infty$ and $\lim_{x\to 3^-} \dfrac{x+4}{x-3} = -\infty$

$\Rightarrow f$ has a vertical asympt $x = 3$.

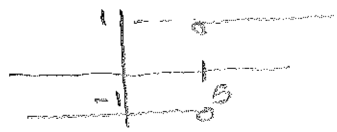
5. $\lim_{x\to 0^+} \left(\dfrac{1}{x} - \ln x\right) = \infty$.

6 (i) $-\infty$ (ii) $-\infty$.

C. TEST QUESTIONS

1. (e) 2 (e)

3. No $\lim_{x\to 5^+} \dfrac{|x+5|}{x-5} = 1$, $\lim_{x\to 5^-} \dfrac{|x+5|}{x-5} = -1$

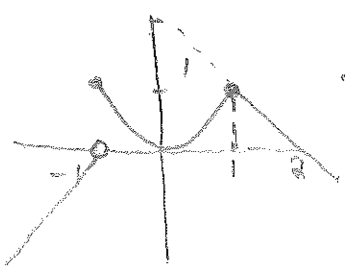


p 92

7. (a) -1 (b) -2 (c) DNE (d) 2 (e) 0 (f) DNE (g) 1 (h) 3

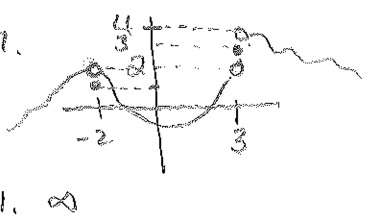
9. (a) $-\infty$ (b) ∞ (c) ∞ (d) $-\infty$ (e) ∞

11.



$\lim_{x\to a} f(x)$ exists for $a \in \mathbb{R} \setminus \{-1\}$

17.



31. ∞

34. $-\infty$

38. $-\infty$